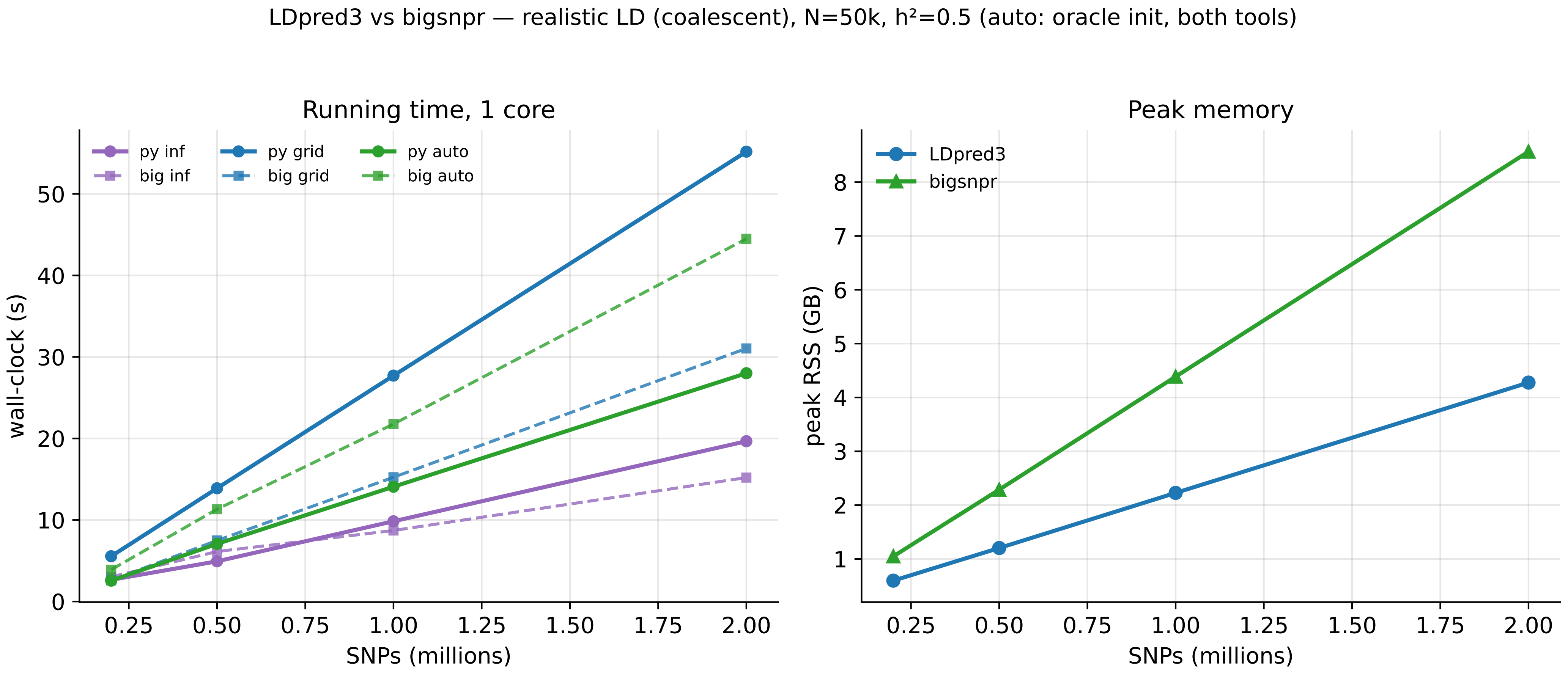
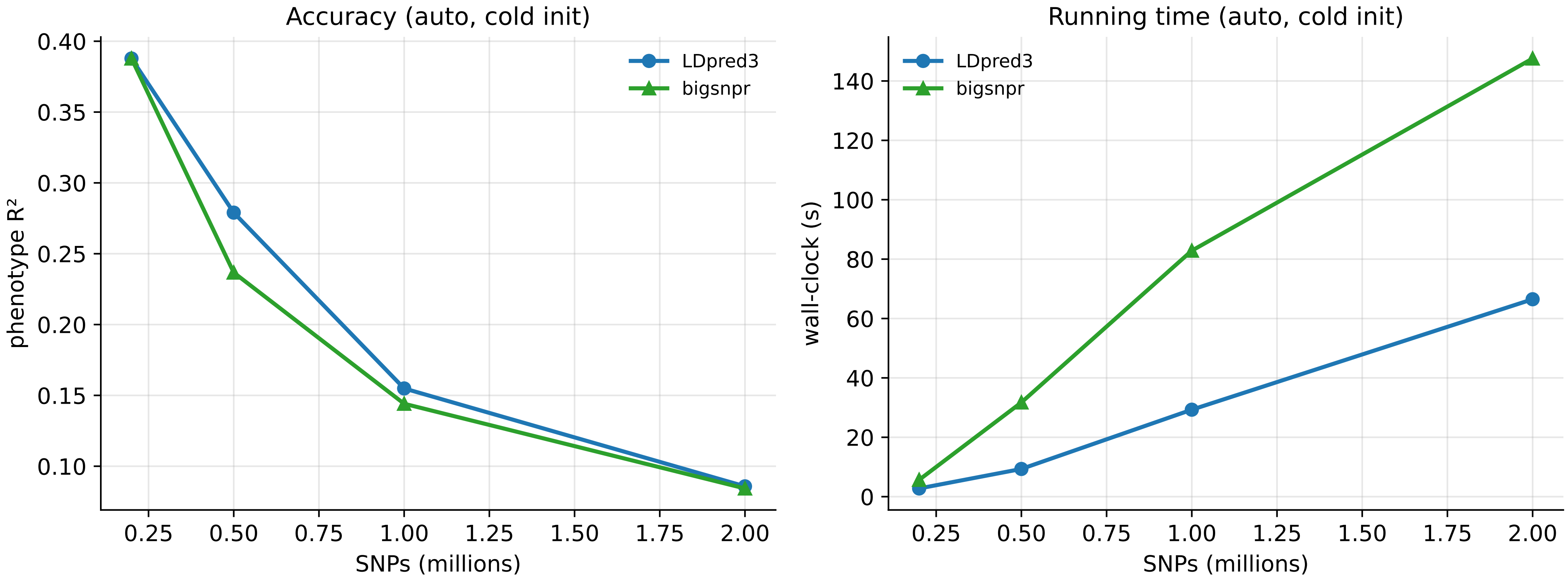
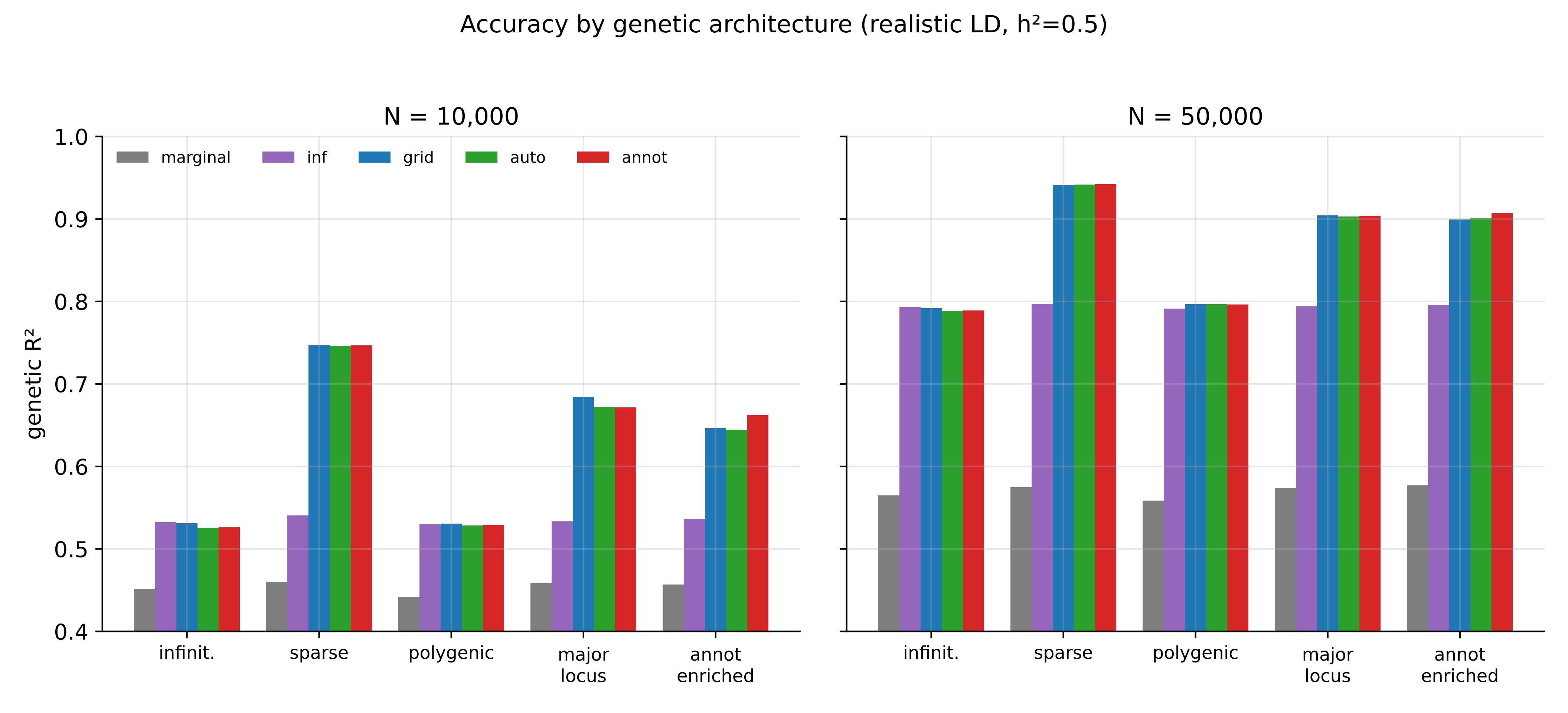
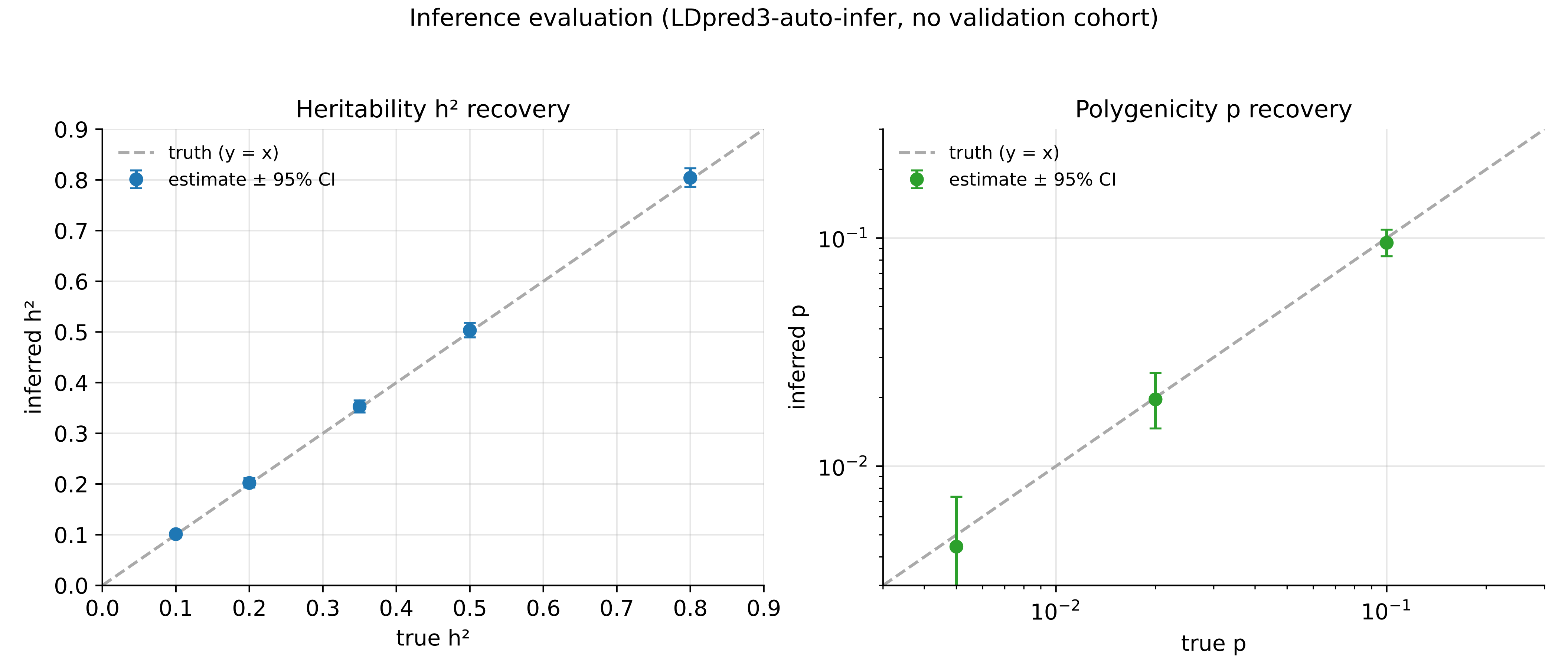

LDpred3 vs bigsnpr — realistic LD (coalescent), N=50k, h²=0.5 (auto: oracle init, both tools)
Running time, 1 core
py inf
big inf
py grid
big grid
py auto
big auto
wall-clock (s)
0
10
20
30
40
50
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
SNPs (millions)
Peak memory
LDpred3
bigsnpr
peak RSS (GB)
1
2
3
4
5
6
7
8
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
SNPs (millions)

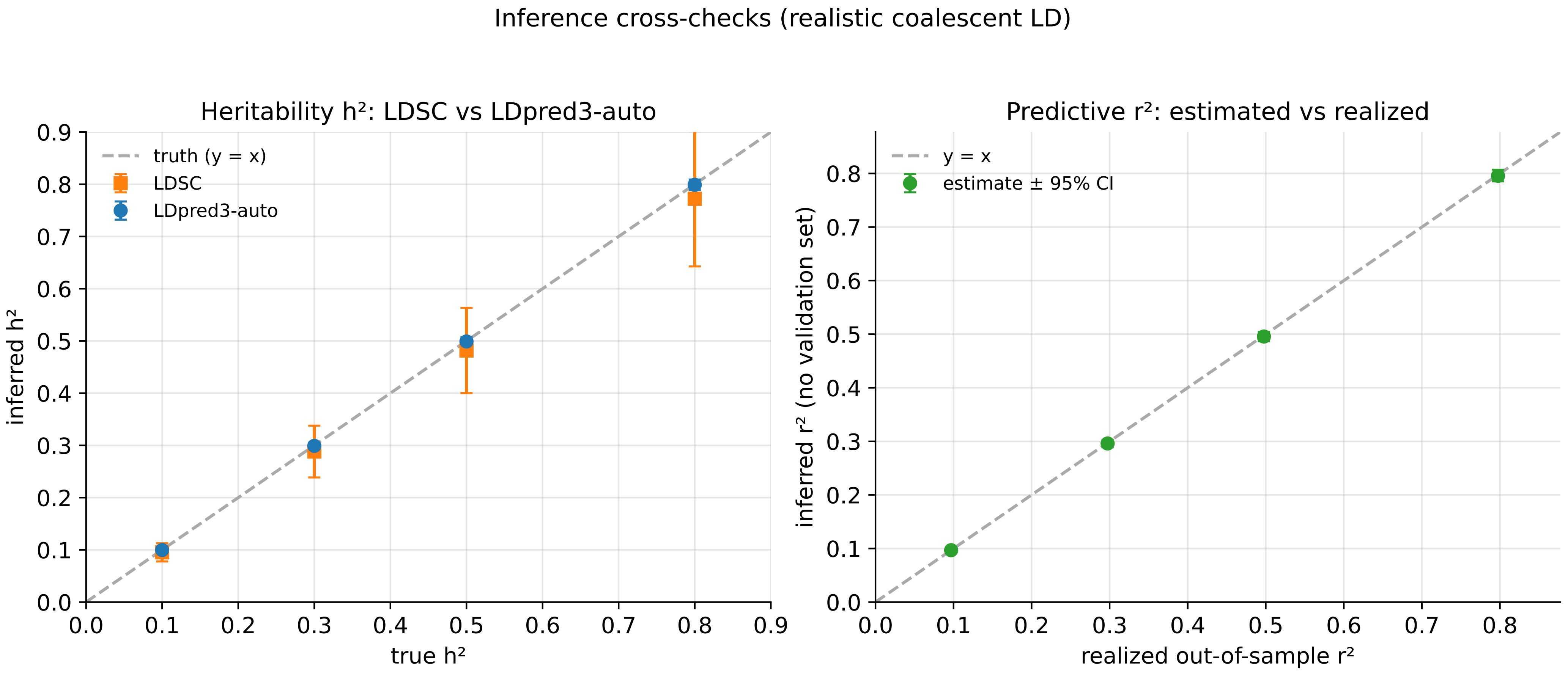
auto from a cold start (h²_init=0.1, p_init=0.1, both tools, single chain)



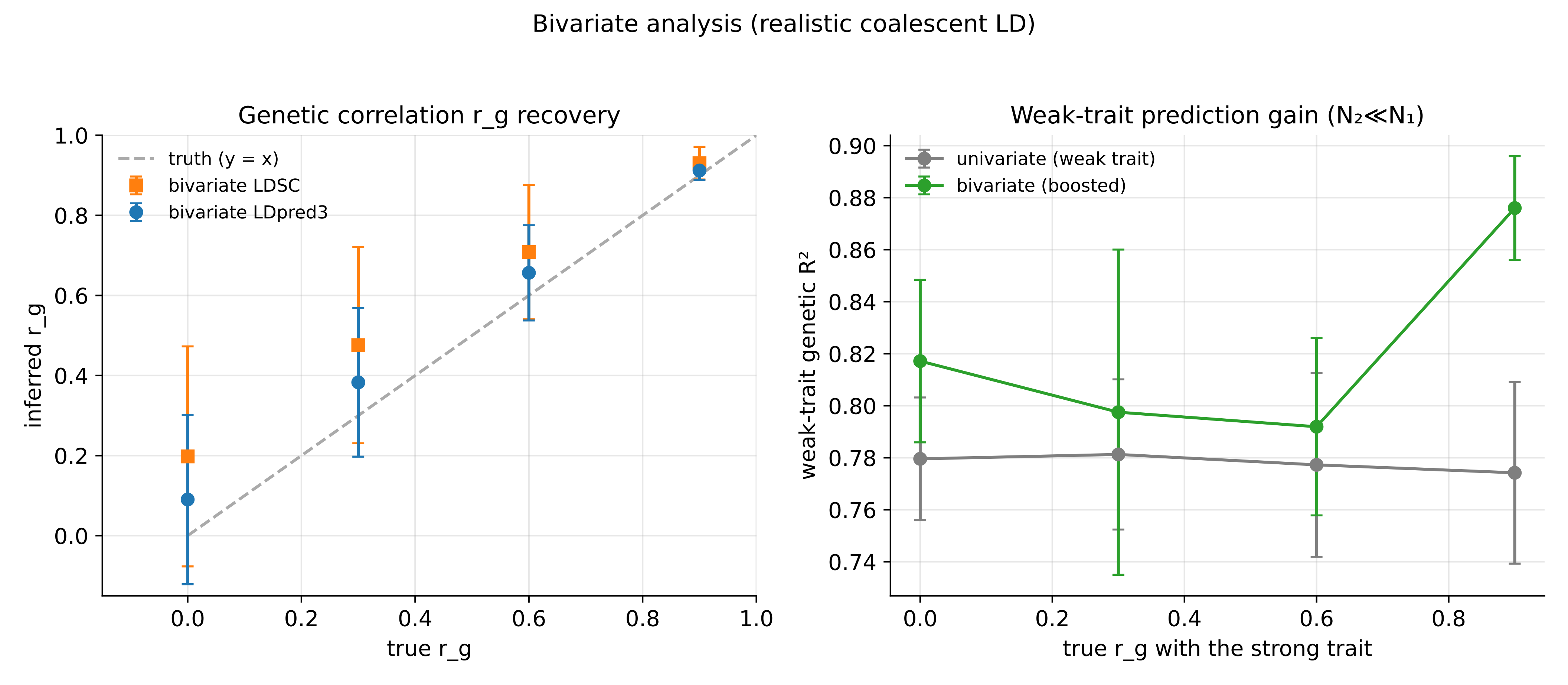


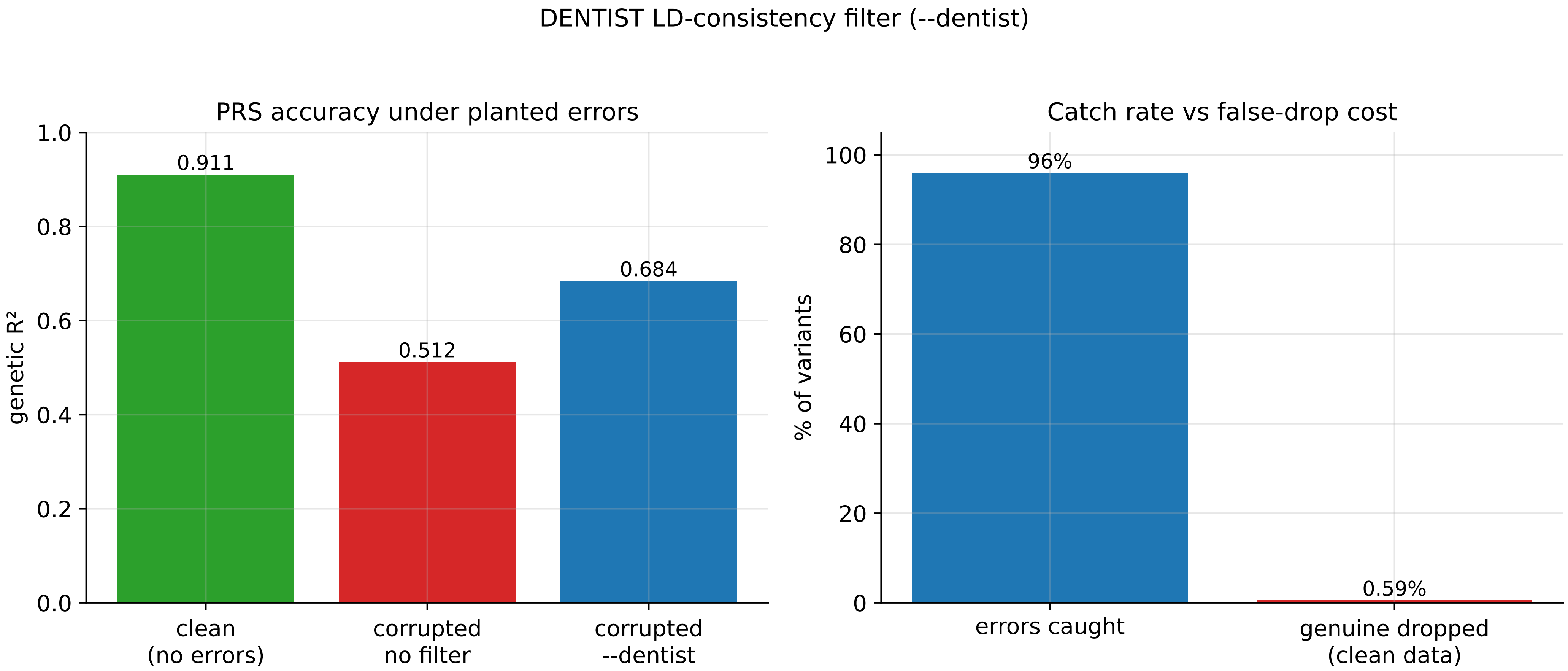
Accuracy by genetic architecture (realistic LD, h²=0.5)
N = 10,000
N = 50,000
marginal
inf
grid
auto
annot
genetic R²
1.0
0.9
0.8
0.7
0.6
0.5
0.4
infinit.
sparse
polygenic
major locus
annot enriched

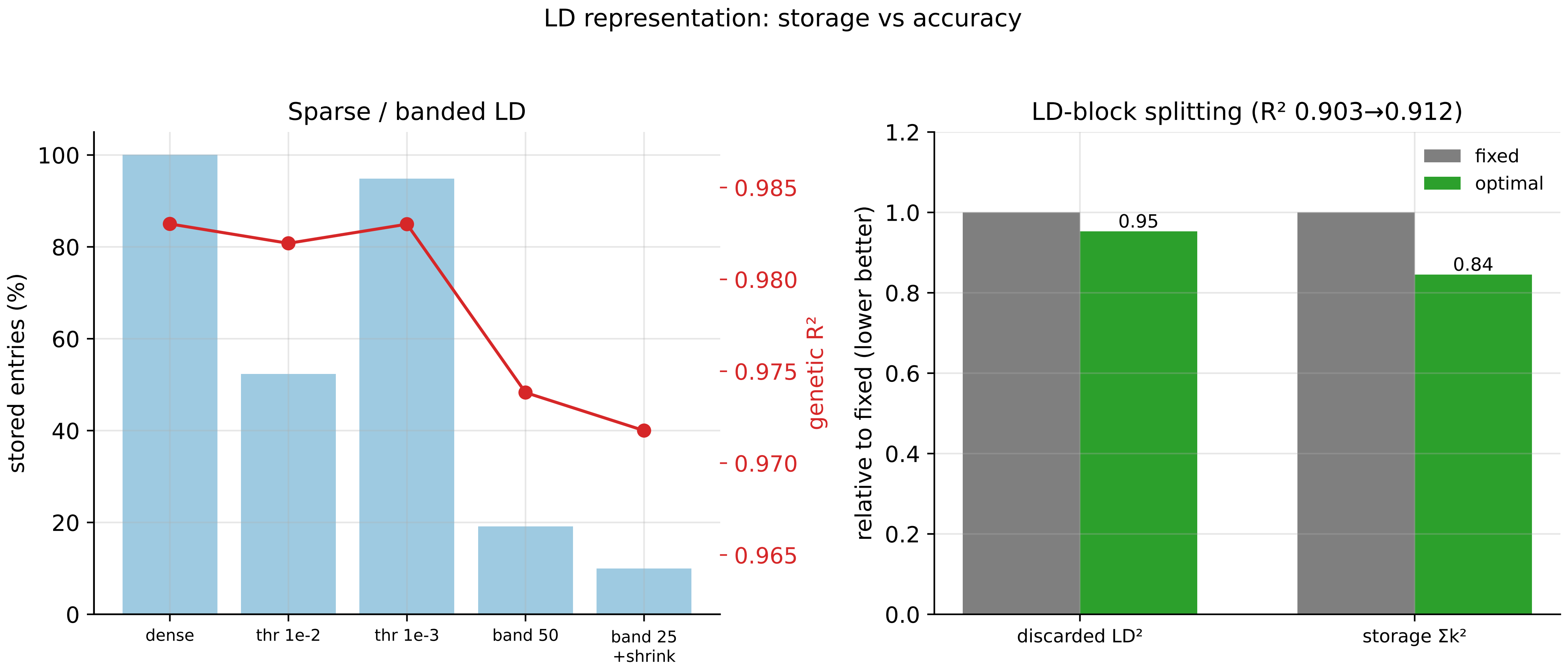

Inference evaluation (LDpred3-auto-infer, no validation cohort)
Heritability h² recovery
truth (y = x)
estimate ± 95% CI
inferred h²
true h²
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Polygenicity p recovery
truth (y = x)
estimate ± 95% CI
inferred p
true p
10^{-1}
10^{-2}



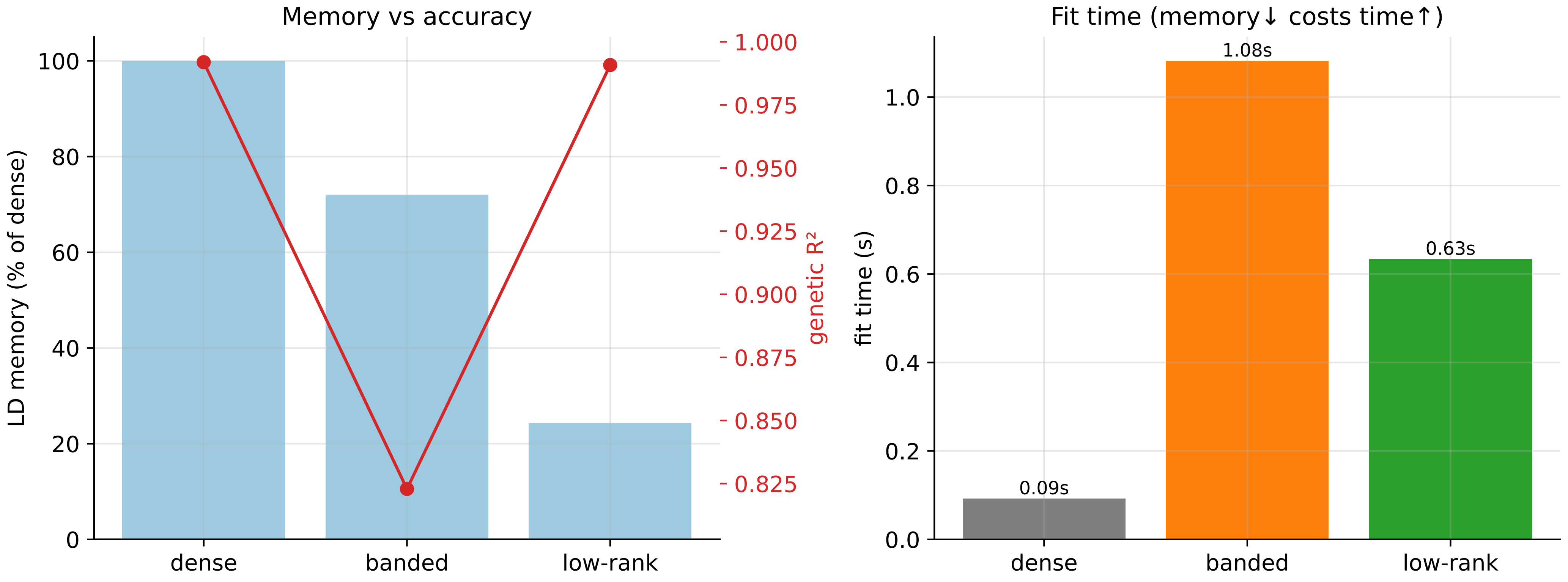
Inference cross-checks (realistic coalescent LD)
Heritability h²: LDSC vs LDpred3-auto
truth (y = x)
LDSC
LDpred3-auto
inferred h²
true h²
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Predictive r²: estimated vs realized
y = x
estimate ± 95% CI
inferred r² (no validation set)
realized out-of-sample r²

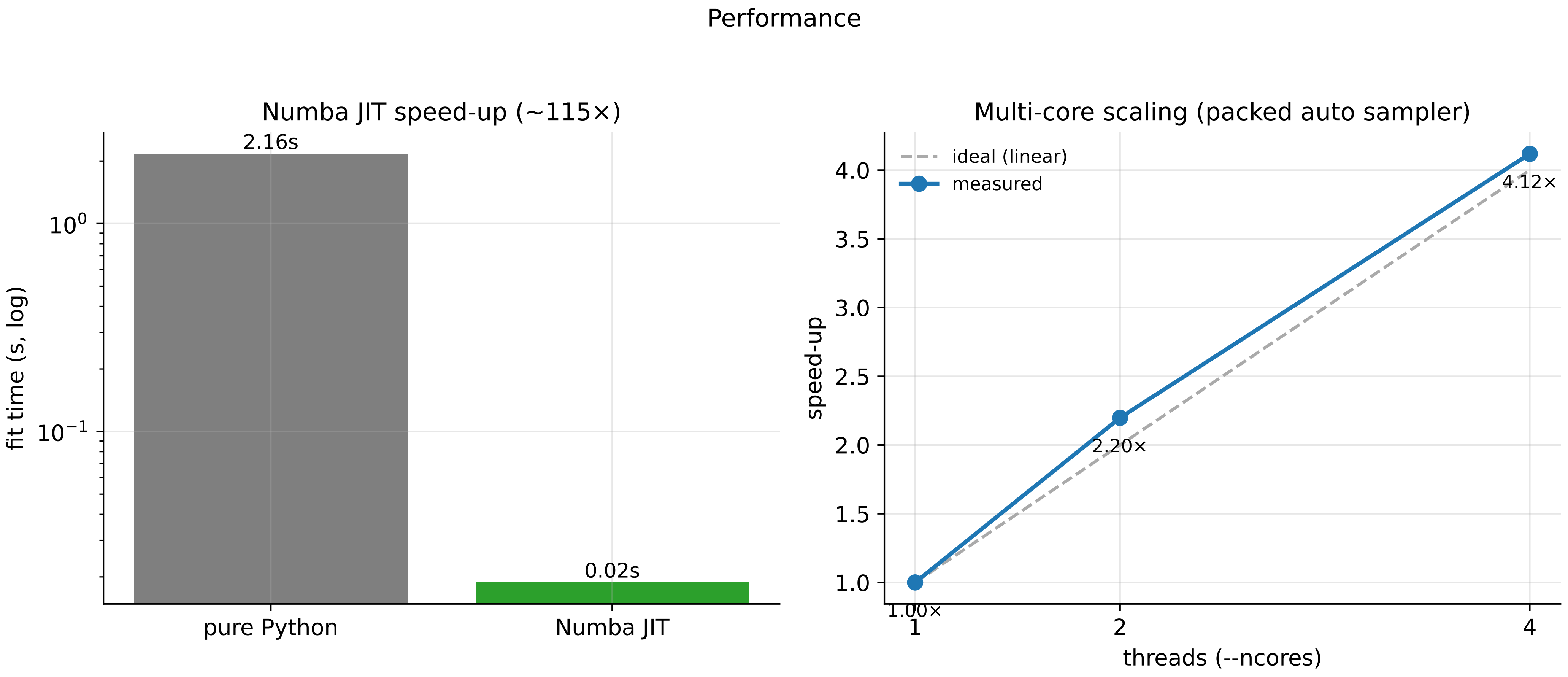

Bivariate analysis (realistic coalescent LD)
Genetic correlation r_g recovery
truth (y = x)
bivariate LDSC
bivariate LDpred3
inferred r_g
true r_g
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Weak-trait prediction gain (N₂≪N₁)
univariate (weak trait)
bivariate (boosted)
weak-trait genetic R²
true r_g with the strong trait
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
0.74
0.0
0.2
0.4
0.6
0.8


DENTIST LD-consistency filter (--dentist)
PRS accuracy under planted errors
genetic R²
1.0
0.8
0.6
0.4
0.2
0.0
0.911
0.512
0.684
clean
(no errors)
corrupted
no filter
corrupted
--dentist
Catch rate vs false-drop cost
% of variants
100
80
60
40
20
0
96%
0.59%
errors caught
genuine dropped
(clean data)


LD representation: storage vs accuracy
Sparse / banded LD
stored entries (%)
100
80
60
40
20
0
dense
thr 1e-2
thr 1e-3
band 50
band 25
+shrink
genetic R²
0.985
0.980
0.975
0.970
0.965
LD-block splitting (R² 0.903→0.912)
relative to fixed (lower better)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
fixed
optimal
0.95
0.84
discarded LD²
storage Σk²

LD representations at scale — realistic coalescent LD (low-rank matches dense at ~1/4 memory; banding loses accuracy)




Performance
Numba JIT speed-up (~115×)
2.16s
0.02s
fit time (s, log)
10^0
10^{-1}
pure Python
Numba JIT
Multi-core scaling (packed auto sampler)
ideal (linear)
measured
4.12×
2.20×
1.00×
speed-up
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1
2
4
threads (--ncores)